Assignment 1

Index-23000422

Report

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Objectives:

The main objective of applying association rule mining to the online retail domain is to discover relationships between products purchased by customers and use these insights to improve business decisions. By identifying frequently co-purchased items, the company can enhance product recommendation systems and provide customers with more relevant suggestions during their shopping experience.The discovered patterns can be used to create effective product bundles and promotional strategies, helping increase sales and customer satisfaction.

The company can also improve inventory management by understanding which products are commonly purchased together and ensuring that related items are available when needed.Furthermore, association rule mining can support better store and website layout decisions by grouping related products together, making it easier for customers to find items of interest. Overall, using these patterns helps the retailer optimize marketing strategies, improve customer experience, reduce unnecessary inventory issues, and increase revenue.

Dataset Description and Preprocessing

The dataset used for this analysis is the Online Retail dataset, which contains transactional data from a UK-based non-store online retailer. The dataset contains transactions recorded between 01/12/2010 and 09/12/2011. Each transaction contains information about purchased products, quantities, invoice details, customer information, and the country of the customer.

The original dataset contains 541,909 records with 8 attributes: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

Before applying association rule mining, several preprocessing steps were performed to transform the raw transactional data into a suitable format for market basket analysis.

Remove missing descriptions

First, missing values in the Description attribute were removed. Product descriptions are required for association rule mining because the algorithm identifies relationships between purchased items. Rows without product descriptions cannot contribute meaningful product relationships. The CustomerID attribute was not used because the objective of this analysis is to identify product purchasing patterns rather than individual customer behaviour.

Remove cancelled invoices

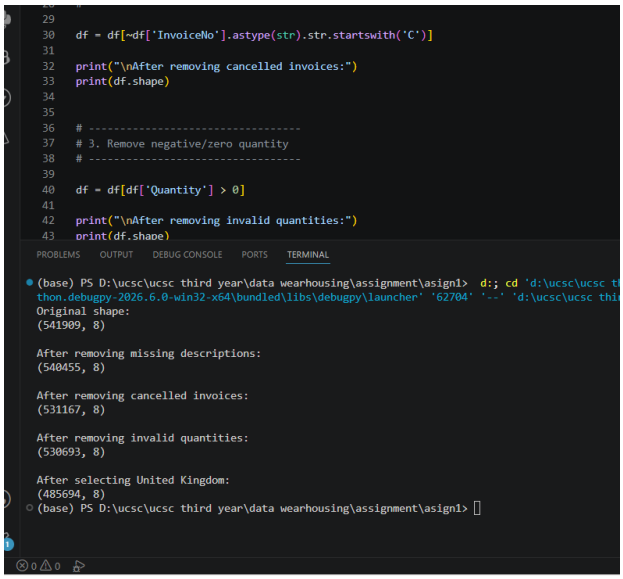
Cancelled transactions were removed by excluding invoices where the InvoiceNo starts with the letter "C". These records represent cancelled or returned orders rather than actual purchases, and including them could introduce incorrect product associations.

Remove negative/zero quantity

Transactions with invalid quantities were also removed by excluding records where Quantity was less than or equal to zero. Negative quantities indicate returned items, while zero quantities do not represent actual purchases. Removing these records ensures that the analysis is based only on completed purchases.s

Select United Kingdom

The Country attribute was used to select a specific region for analysis. The United Kingdom was selected because it contains the highest number of transactions in the dataset (495,478 records), providing a larger and more reliable set of transactions for discovering purchasing patterns.



Attribute Selection

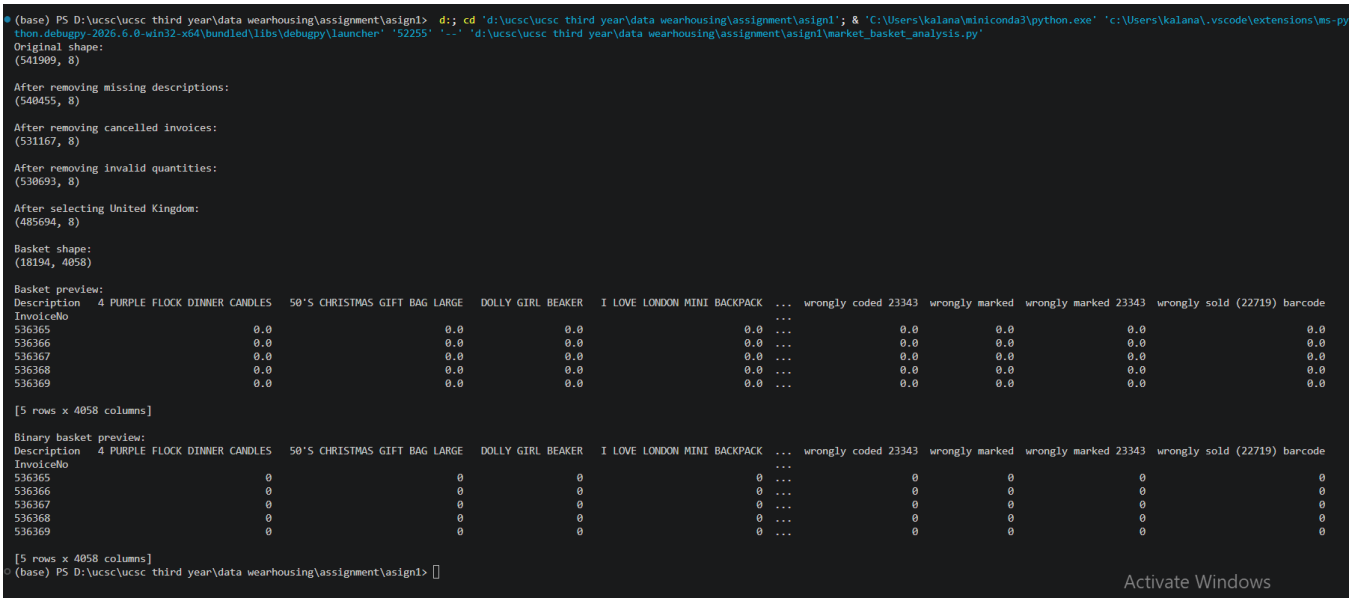
The attributes UnitPrice and InvoiceDate were not considered in the association rule mining process. UnitPrice was ignored because the objective is to find relationships between products rather than analyze pricing behaviour. InvoiceDate was not used because time-based purchasing trends were outside the scope of this market basket analysis.

After preprocessing, the remaining transaction data will be converted into a basket format where each invoice represents a customer's basket and each product represents an item that can be analyzed for associations.

Transaction Transformation

After preprocessing, the transactional dataset was transformed into a **basket (transaction) format** suitable for association rule mining. Each unique **InvoiceNo** represents a single transaction, while each **Description** represents a product. The quantity values were converted into binary values, where **1** indicates that the product was purchased in the transaction and **0** indicates that it was not purchased.

Finally, the cleaned transactional data was transformed into a basket matrix containing **18,194 transactions** and **4,058 unique products**, which was used as the input for the Apriori association rule mining algorithm.



```
(base) PS D:\ucsc\ucsc third year\data wearhousing\assignment\asign1>  d:; cd 'd:\ucsc\ucsc third year\data wearhousing\assignment\asign1'; & 'C:\Users\kalana\miniconda3\python.exe' 'c:\Users\kalana\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '52255' '--' 'd:\ucsc\ucsc third year\data wearhousing\assignment\asign1\market_basket_analysis.py'
Original shape:
(541909, 8)

After removing missing descriptions:
(540455, 8)

After removing cancelled invoices:
(531167, 8)

After removing invalid quantities:
(530693, 8)

After selecting United Kingdom:
(485694, 8)

Basket shape:
(18194, 4058)

Basket preview:
Description   4 PURPLE FLOCK DINNER CANDLES   50'S CHRISTMAS GIFT BAG LARGE   DOLLY GIRL BEAKER   I LOVE LONDON MINI BACKPACK  ...  wrongly coded 23343  wrongly marked  wrongly marked 23343  wrongly sold (22719) barcode
InvoiceNo                                                                                                                  ...
536365                                  0.0                             0.0              0.0                          0.0  ...                  0.0            0.0                   0.0                          0.0
536366                                  0.0                             0.0              0.0                          0.0  ...                  0.0            0.0                   0.0                          0.0
536367                                  0.0                             0.0              0.0                          0.0  ...                  0.0            0.0                   0.0                          0.0
536368                                  0.0                             0.0              0.0                          0.0  ...                  0.0            0.0                   0.0                          0.0
536369                                  0.0                             0.0              0.0                          0.0  ...                  0.0            0.0                   0.0                          0.0

[5 rows x 4058 columns]

Binary basket preview:
Description   4 PURPLE FLOCK DINNER CANDLES   50'S CHRISTMAS GIFT BAG LARGE   DOLLY GIRL BEAKER   I LOVE LONDON MINI BACKPACK  ...  wrongly coded 23343  wrongly marked  wrongly marked 23343  wrongly sold (22719) barcode
InvoiceNo                                                                                                                  ...
536365                                  0                               0                0                            0  ...                    0              0                     0                            0
536366                                  0                               0                0                            0  ...                    0              0                     0                            0
536367                                  0                               0                0                            0  ...                    0              0                     0                            0
536368                                  0                               0                0                            0  ...                    0              0                     0                            0
536369                                  0                               0                0                            0  ...                    0              0                     0                            0

[5 rows x 4058 columns]
(base) PS D:\ucsc\ucsc third year\data wearhousing\assignment\asign1>
```

Activate Windows

Rule mining process:

Choice of Algorithm

The Apriori algorithm was selected to perform association rule mining because it is a widely used algorithm for discovering frequent itemsets and generating association rules from transactional datasets. The algorithm identifies product combinations that occur frequently together by using a minimum support threshold and then generates association rules based on a minimum confidence threshold. Since the Online Retail dataset consists of customer transactions, Apriori is well suited for identifying purchasing patterns and product associations.

Parameter Settings

Parameter	Value(s) Tested	Final Value
Algorithm	Apriori	Apriori
Minimum Support	0.05, 0.03, 0.02	0.02
Minimum Confidence	0.5	0.5
Rule Evaluation metric	Lift	Lift

The confidence threshold was fixed at **0.50**, meaning that only rules with at least **50% confidence** were considered. The lift metric was used to rank the generated rules because it measures the strength of the relationship between products beyond what would be expected by chance.

Experimental Process

Three experiments were conducted to observe the effect of different minimum support values on the number of frequent itemsets, generated association rules, and execution time.

Experiment	Minimum Support	Minimum Confidence	Execution Time	Frequent Itemsets	Rules
1	0.05	0.5	0.16	33	0
2	0.03	0.5	1.77	140	13
3	0.02	0.5	8.92	292	72

Purpose of Repeated Experiments

The experiments were repeated using different minimum support values to determine how the support threshold influences the number of frequent itemsets, association rules, and computational cost. A higher support threshold filters out less frequent product combinations, resulting in fewer patterns but faster execution. Lowering the support threshold allows more product combinations to be discovered, although it increases the processing time.

Feedback from Previous Cycles

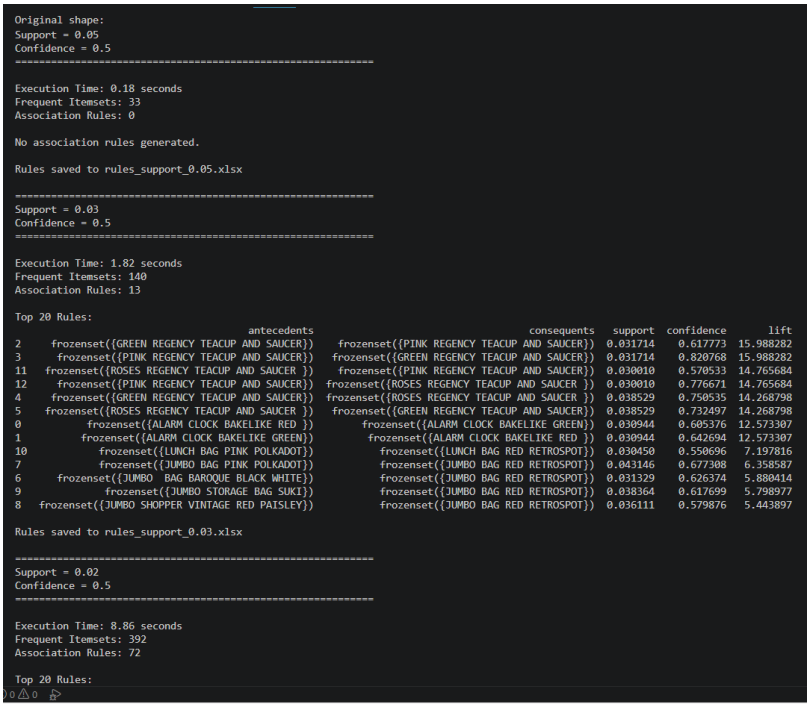
In the first experiment, a minimum support value of 0.05 produced only 33 frequent itemsets and no association rules because the support threshold was too restrictive. Although the execution time was only 0.16 seconds, the discovered patterns were insufficient for meaningful analysis.

The support threshold was then reduced to 0.03. This increased the number of frequent itemsets to 140 and generated 13 association rules. The execution time also increased to 1.77 seconds. While the results were more informative than the first experiment, the number of discovered rules remained relatively limited.

Finally, the minimum support was reduced to 0.02. This produced 392 frequent itemsets and 72 association rules, providing a much richer set of product associations. The execution time increased to 8.92 seconds because the algorithm evaluated a larger number of candidate itemsets. Despite the additional computation time, this parameter setting generated the most useful and comprehensive results.

Final Parameter Selection

The final model used a minimum support of 0.02 and a minimum confidence of 0.50. These values produced the highest number of meaningful association rules while maintaining acceptable execution time. Therefore, this parameter setting was selected for the final analysis presented in the report.



```
Original shape:
Support = 0.05
Confidence = 0.5
============================================================

Execution Time: 0.18 seconds
Frequent Itemsets: 33
Association Rules: 0

No association rules generated.

Rules saved to rules_support_0.05.xlsx

============================================================
Support = 0.03
Confidence = 0.5
============================================================

Execution Time: 1.82 seconds
Frequent Itemsets: 140
Association Rules: 13

Top 20 Rules:
                                      antecedents                                      consequents   support  confidence       lift
2     frozenset({GREEN REGENCY TEACUP AND SAUCER})     frozenset({PINK REGENCY TEACUP AND SAUCER})  0.031714    0.617773  15.988282
3      frozenset({PINK REGENCY TEACUP AND SAUCER})    frozenset({GREEN REGENCY TEACUP AND SAUCER})  0.031714    0.820768  15.988282
11   frozenset({ROSES REGENCY TEACUP AND SAUCER })     frozenset({PINK REGENCY TEACUP AND SAUCER})  0.030010    0.570533  14.765684
12     frozenset({PINK REGENCY TEACUP AND SAUCER})   frozenset({ROSES REGENCY TEACUP AND SAUCER })  0.030010    0.776671  14.765684
4     frozenset({GREEN REGENCY TEACUP AND SAUCER})   frozenset({ROSES REGENCY TEACUP AND SAUCER })  0.038529    0.750535  14.268798
5    frozenset({ROSES REGENCY TEACUP AND SAUCER })    frozenset({GREEN REGENCY TEACUP AND SAUCER})  0.038529    0.732497  14.268798
0           frozenset({ALARM CLOCK BAKELIKE RED })         frozenset({ALARM CLOCK BAKELIKE GREEN})  0.030944    0.605376  12.573307
1         frozenset({ALARM CLOCK BAKELIKE GREEN})          frozenset({ALARM CLOCK BAKELIKE RED })  0.030944    0.642694  12.573307
10             frozenset({LUNCH BAG PINK POLKADOT})            frozenset({LUNCH BAG RED RETROSPOT})  0.030450    0.550696   7.197816
7              frozenset({JUMBO BAG PINK POLKADOT})            frozenset({JUMBO BAG RED RETROSPOT})  0.043146    0.677308   6.358587
6      frozenset({JUMBO  BAG BAROQUE BLACK WHITE})            frozenset({JUMBO BAG RED RETROSPOT})  0.031329    0.626374   5.880414
9            frozenset({JUMBO STORAGE BAG SUKI})            frozenset({JUMBO BAG RED RETROSPOT})  0.038364    0.617699   5.798977
8   frozenset({JUMBO SHOPPER VINTAGE RED PAISLEY})          frozenset({JUMBO BAG RED RETROSPOT})  0.036111    0.579876   5.443897

Rules saved to rules_support_0.03.xlsx

============================================================
Support = 0.02
Confidence = 0.5
============================================================

Execution Time: 8.86 seconds
Frequent Itemsets: 392
Association Rules: 72

Top 20 Rules:
```

Resulting Rules

Summary of Generated Rules

The final Apriori model was executed using a minimum support of **0.02** and a minimum confidence of **0.50**. This configuration generated **392 frequent itemsets** and **72 association rules**. The discovered rules reveal purchasing relationships between products that customers frequently buy together in the same transaction.

Many of the rules involve products from the same product line or complementary items, such as tea cup sets, lunch boxes, Christmas decorations, shopping bags, and gardening products. Several rules achieved lift values greater than 10, indicating strong associations that are significantly more likely than random co-occurrence. These findings provide useful insights for marketing, cross-selling, product placement, and inventory planning.

Selected Rules

Rule 1

WOODEN STAR CHRISTMAS SCANDINAVIAN → WOODEN HEART CHRISTMAS SCANDINAVIAN

- Support: **2.02%**
- Confidence: **76.83%**
- Lift: **27.46**

This rule indicates that customers who purchase the Wooden Star Christmas Scandinavian decoration are highly likely to also purchase the Wooden Heart Christmas Scandinavian decoration. The very high lift value shows an exceptionally strong relationship, making these products ideal candidates for bundle promotions or joint displays during the Christmas season.

Rule 2

GREEN REGENCY TEACUP AND SAUCER → PINK REGENCY TEACUP AND SAUCER

- Support: **3.17%**
- Confidence: **61.78%**
- Lift: **15.99**

Customers who purchase the green Regency teacup and saucer frequently purchase the pink version in the same transaction. This suggests that customers prefer buying matching tea set collections rather than individual pieces.

Rule 3

DOLLY GIRL LUNCH BOX → SPACEBOY LUNCH BOX

- Support: **2.34%**
- Confidence: **60.94%**
- Lift: **15.82**

This rule shows that customers who buy the Dolly Girl Lunch Box often purchase the Spaceboy Lunch Box as well. Since these products appeal to similar customer groups, they are suitable for cross-selling and promotional offers.

Rule 4

STRAWBERRY CHARLOTTE BAG → WOODLAND CHARLOTTE BAG

- Support: **2.06%**
- Confidence: **55.57%**
- Lift: **14.80**

Customers purchasing the Strawberry Charlotte Bag also tend to purchase the Woodland Charlotte Bag. This indicates that shoppers often buy multiple designs from the same product range, suggesting an opportunity to market these bags as a collection.

Rule 5

GARDENERS KNEELING PAD KEEP CALM → GARDENERS KNEELING PAD CUP OF TEA

- Support: **2.97%**
- Confidence: **60.24%**
- Lift: **14.61**

Customers who purchase one gardening kneeling pad design frequently purchase another design in the same transaction. This behaviour suggests interest in themed gardening products or purchasing multiple items as gifts.

Rule 6

JUMBO BAG PINK POLKADOT → JUMBO BAG RED RETROSPOT

- Support: **4.31%**
- Confidence: **67.73%**
- Lift: **6.36**

Customers who purchase the Jumbo Bag Pink Polkadot are also likely to purchase the Jumbo Bag Red Retrospot. Although the lift is lower than the previous rules, this rule has the highest support among the selected rules, indicating that it occurs relatively frequently and is valuable for product recommendations and shelf placement.

Recommendations

The discovered association rules provide valuable insights into customer purchasing behaviour and can be used to improve sales, marketing strategies, and inventory management. Based on the results, the following recommendations are proposed.

1. Introduce Product Bundles

Products with strong associations, such as the **Wooden Star Christmas Scandinavian** and **Wooden Heart Christmas Scandinavian** decorations, or the **Green Regency Teacup and Saucer** and **Pink Regency Teacup and Saucer**, should be offered as bundled products. Selling these items together can encourage customers to purchase multiple products in a single transaction and increase the average order value.

2. Implement Cross-Selling Recommendations

When customers add a product to their shopping cart, the system should recommend related products with high confidence. For example, customers purchasing the **Dolly Girl Lunch Box** can be shown the **Spaceboy Lunch Box**, while customers selecting the **Jumbo Bag Pink Polkadot** can be recommended the **Jumbo Bag Red Retrospot**. This can improve customer convenience while increasing sales.

3. Improve Product Placement

Frequently purchased products should be displayed together on the online store. Related products can be placed within the same category page or displayed under sections such as "Frequently Bought Together" or "Customers Also Purchased." This makes it easier for customers to discover complementary products.

4. Optimize Inventory Management

Products that frequently appear together should be stocked together to reduce the risk of one product being unavailable while its associated product is in stock. Maintaining sufficient inventory for strongly associated products helps prevent missed sales opportunities and improves customer satisfaction.

5. Design Seasonal Promotions

Several strong associations involve Christmas-themed products. These products can be promoted together during the holiday season through special discounts, bundle offers, and targeted marketing campaigns to increase seasonal sales.

6. Support Future Business Decisions

The association rules should be monitored and updated periodically as customer purchasing behaviour changes over time. Repeating the analysis on new transaction data will help the company identify emerging buying patterns and continuously improve product recommendations, promotional strategies, and inventory planning.